A PROJECT Title

ON

NextGen Travel Intelligence and Automation Portal

Insem-End project submitted in partial fulfilment of the requirements for the Course

Front End Development Frameworks ***for the degree of***

BACHELOR OF TECHNOLOGY

IN

Computer Science and Engineering

(2025-2026 A.Y)

BY

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Name of Title: NextGen Travel Intelligence and Automation Portal

1. Abstract:

The NextGen Travel Intelligence and Automation Portal is designed to make corporate travel simpler, faster, and more organized. In many organizations, managing employee travel involves multiple steps like planning trips, booking flights, getting approvals, and tracking expenses, which can be time-consuming and complex. This system brings all these activities into one easy-to-use platform, helping companies handle travel more efficiently while reducing manual work and errors.

The portal allows administrators to set clear travel rules such as budget limits, preferred airlines, and booking conditions, ensuring that every trip follows company policies. Employees can easily request travel through a simple interface, and managers can quickly approve or reject requests in real time. To make the system more advanced and practical, features like smart flight suggestions, price alerts, fraud detection, travel history, notifications, and role-based dashboards can be included. In addition, the system can offer options for in-flight food selection, provide detailed information about the pilot and crew, and allow users to compare different flights based on company policies, time, and cost. It can also include live flight status tracking, digital boarding passes for a paperless experience, and multi-language support to improve accessibility for users from different regions. Additional functionalities such as emergency support, payment integration, and mobile access make the system more flexible and user-friendly.

Finally, the proposed portal delivers a complete and efficient solution for managing corporate travel in a smart and organized manner. It not only helps in tracking travel expenses and maintaining all records in a centralized system, but also ensures better transparency and control over employee travel activities. The inclusion of user-friendly dashboards and basic analytics makes it easier for organizations to understand travel patterns and manage budgets effectively. Overall, this project aims to simplify complex travel processes by combining automation, policy control, and data management, resulting in improved efficiency, reduced costs, and a more convenient travel experience for both employees and management.

2. INTRODUCTION:

The NextGen Travel Intelligence and Automation Portal is a smart web-based application developed to simplify and automate travel management activities. In many organizations, handling travel requests, flight bookings, approvals, and travel information manually can be time-consuming and difficult to manage. This project provides a centralized platform where users can access travel-related services in a simple, organized, and user-friendly manner.

The portal is designed using modern frontend technologies such as ReactJS, HTML, CSS, and JavaScript to provide an interactive and responsive user experience. Users can search flights, access travel details, view digital boarding passes, and navigate through different travel services easily through a clean interface. The application also includes features such as live flight status display, smart navigation, and user-friendly booking forms to improve convenience and usability.

The main objective of this project is to develop an efficient and attractive travel management interface that reduces complexity and improves the overall user experience. By combining automation, responsive design, and organized data presentation, the system provides a modern solution for handling travel-related activities effectively and conveniently

3. SOFTWARE REQUIREMENTS:

Software Requirements Specification

Frontend Technologies

- HTML5
- CSS3
- JavaScript
- ReactJS

Development Tools

- Visual Studio Code
- Git & GitHub
- Postman (optional)

Operating System

- Windows 10 / Windows 11

Browser Support

- Google Chrome
- Microsoft Edge

Additional Libraries and Frameworks

- React Router
- Bootstrap / Tailwind CSS

Internet Requirements

- Stable internet connection for smooth application usage and testing.

Functional Requirements

- User Registration Interface
- Flight Search Module
- Booking Form
- Travel Information Display
- Digital Boarding Pass
- User-friendly Navigation

Non-Functional Requirements

- Attractive User Interface
- Fast Response Time
- Easy Navigation
- Reliability
- Maintainability

Course Coordinator	**Course Instructor**	**FED , Coordinator**